

Tasks@DS\Ecommerce.py

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1  # Mini project-2
2
3
4  # Ecommerce project using Numpy, Pandas, and Matplotlib
5
6  # USING PANDAS
7  import pandas as pd
8
9  # USING NUMPY
10 import numpy as np
11
12 # USING MATPLOTLIB
13 import matplotlib.pyplot as plt
14
15 # DUMMY DATA
16 data = {
17     "Order_ID": [101,102,103,104,105,106,107,108,109,110,
18                 111,112,113,114,115,116,117,118,119,120,
19                 121,122,123,124,125,126,127,128,129,130],
20
21     "Product": ["Laptop", "Mobile", "Shoes", "TShirt", "Headphones",
22                "Watch", "Laptop", "Shoes", "Mobile", "Watch",
23                "TShirt", "Headphones", "Laptop", "Mobile", "Shoes",
24                "Watch", "Laptop", "Headphones", "TShirt", "Mobile",
25                "Shoes", "Laptop", "Watch", "Headphones", "Mobile",
26                "TShirt", "Shoes", "Laptop", "Watch", "Mobile"],
27
28     "Category": ["Electronics", "Electronics", "Fashion", "Fashion", "Electronics",
29                 "Accessories", "Electronics", "Fashion", "Electronics", "Accessories",
30                 "Fashion", "Electronics", "Electronics", "Electronics", "Fashion",
31                 "Accessories", "Electronics", "Electronics", "Fashion", "Electronics",
32                 "Fashion", "Electronics", "Accessories", "Electronics", "Electronics",
33                 "Fashion", "Fashion", "Electronics", "Accessories", "Electronics"],
34
35     "Quantity": [2,3,4,5,6,2,1,3,2,4,
36                 6,5,2,4,3,5,1,6,4,2,
37                 5,2,3,4,2,6,3,1,5,4],
38
39     "Price": [50000,20000,2500,1000,1500,
40              3000,50000,2500,20000,3000,
41              1000,1500,50000,20000,2500,
42              3000,50000,1500,1000,20000,
43              2500,50000,3000,1500,20000,
44              1000,2500,50000,3000,20000],
45
46     "City": ["Jaipur", "Delhi", "Mumbai", "Pune", "Bangalore",
47             "Jaipur", "Delhi", "Mumbai", "Pune", "Bangalore",
48             "Jaipur", "Delhi", "Mumbai", "Pune", "Bangalore",

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49         "Jaipur", "Delhi", "Mumbai", "Pune", "Bangalore",
50         "Jaipur", "Delhi", "Mumbai", "Pune", "Bangalore",
51         "Jaipur", "Delhi", "Mumbai", "Pune", "Bangalore"],
52
53     "Month": ["Jan", "Jan", "Feb", "Feb", "Mar",
54              "Mar", "Apr", "Apr", "May", "May",
55              "Jun", "Jun", "Jul", "Jul", "Aug",
56              "Aug", "Sep", "Sep", "Oct", "Oct",
57              "Nov", "Nov", "Dec", "Dec", "Jan",
58              "Feb", "Mar", "Apr", "May", "Jun"]
59 }
60
61 df = pd.DataFrame(data)
62
63 # CONVERTING INTO CSV FILE
64 df.to_csv("ecommerce_sales.csv", index=False)
65
66 # READING CSV FILE
67 df = pd.read_csv("ecommerce_sales.csv")
68
69 # ADDING SALES COLUMN
70 df["Sales"] = df["Quantity"] * df["Price"]
71
72 # CALCULATE TOTAL REVENUE
73 total_sales = df["Sales"].sum()
74
75 # AVERAGE SALES VALUE
76 average_sales = np.mean(df["Sales"])
77
78 # HIGHEST SALE
79 highest_sale = np.max(df["Sales"])
80
81 # LOWEST SALE
82 lowest_sale = np.min(df["Sales"])
83
84 # BEST SELLING PRODUCT
85 best_selling = df.groupby("Product")["Quantity"].sum()
86
87 # TOP PRODUCT BY REVENUE
88 product_sales = df.groupby("Product")["Sales"].sum()
89
90 # FIND THE HIGHEST REVENUE PRODUCT
91 top_product = product_sales.idxmax()
92
93 # CITY-WISE SALES ANALYSIS
94 city_sales = df.groupby("City")["Sales"].sum()
95
96 # CATEGORY-WISE SALES ANALYSIS
97 category_sales = df.groupby("Category")["Sales"].sum()
98
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99 # TOP CATEGORY BY REVENUE
100 category_revenue = category_sales.idxmax()
101
102 # FIND THE TOP CITY
103 top_city = city_sales.idxmax()
104
105 # MONTHLY SALES TREND
106 month_order = ["Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]
107 monthly_sales = df.groupby("Month")["Sales"].sum()
108 monthly_sales = monthly_sales.reindex(month_order)
109
110 def sales_summary():
111     print("\n=== SALES SUMMARY ===")
112     print("Total Revenue =", total_sales)
113     print("Average Sale value =", average_sales)
114     print("Highest Sale =", highest_sale)
115     print("Lowest sale =", lowest_sale)
116
117
118 def product_analysis():
119     print("\n=== PRODUCT ANALYSIS ===")
120     print(product_sales)
121     print("\nHighest Revenue Product =", top_product)
122     print("\nBest Selling Product =", best_selling.idxmax())
123
124
125 def city_analysis():
126     print("\n=== CITY ANALYSIS ===")
127     print(city_sales)
128     print("\nTop Performing City =", top_city)
129
130
131 def monthly_analysis():
132     print("\n=== MONTHLY SALES ANALYSIS ===")
133     print(monthly_sales)
134
135
136
137 def category_analysis():
138     print("\n=== CATEGORY ANALYSIS ===")
139     print(category_sales)
140     print("\nTop Category by Revenue =", category_revenue)
141
142
143 def show_charts():
144
145     plt.figure(figsize=(14,10))
146
147 # BAR CHART
148     plt.subplot(2,2,1)
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```
149     product_sales.plot(kind="bar")
150     plt.title("Revenue by Product")
151
152     # PIE CHART
153     plt.subplot(2,2,2)
154     category_sales.plot(kind="pie", autopct="%1.1f%%")
155     plt.title("Category Revenue")
156     plt.ylabel("")
157
158     # LINE CHART
159     plt.subplot(2,2,3)
160     monthly_sales.plot(kind="line", marker="o")
161     plt.title("Monthly Sales Trend")
162
163     # HISTOGRAM
164     plt.subplot(2,2,4)
165     plt.hist(df["Sales"])
166     plt.title("Sales Distribution")
167
168     plt.tight_layout()
169     plt.show()
170
171
172     while True:
173         print("\n===== E-COMMERCE SALES ANALYSIS =====")
174         print("1. Sales Summary")
175         print("2. Product Analysis")
176         print("3. City Analysis")
177         print("4. Category Analysis")
178         print("5. Monthly Analysis")
179         print("6. Show Charts")
180         print("7. Exit")
181
182         choice = int(input("Enter Choice: "))
183
184         if choice == 1:
185             sales_summary()
186
187         elif choice == 2:
188             product_analysis()
189
190         elif choice == 3:
191             city_analysis()
192
193         elif choice == 4:
194             category_analysis()
195
196         elif choice == 5:
197             monthly_analysis()
198
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```
199     elif choice == 6:
200         show_charts()
201
202     elif choice == 7:
203         print("Thank You!")
204         break
205
206     else:
207         print("Invalid Choice!")
```